VLAN and Inter-VLAN Routing Simulation Project

A guided project with real deliverables

Scenario

You're the network engineer for a small company, "NovaTech", with 3 floors, 4 departments, and a guest Wi-Fi network. You design, build, secure, test, and document the network in a simulator.

- **Tools:** Cisco Packet Tracer (easiest), or GNS3 / EVE-NG if you want real IOS images
- **Time:** 10-15 hours over 1-2 weeks
- **Level:** Beginner to intermediate (CCNA-style)

Topology

- 1 router (router-on-a-stick), later upgraded to a Layer 3 switch (SVIs)
- 1 core/distribution switch + 2 access switches
- 8-10 PCs, 1 server, 1 wireless AP (guest)
- Optional: second router to practice static and OSPF routing

VLAN and IP Plan

VLAN	Name	Subnet	Gateway
10	Sales	192.168.10.0/24	.1
20	Engineering	192.168.20.0/24	.1
30	HR	192.168.30.0/24	.1
40	Servers	192.168.40.0/24	.1
50	Guest	192.168.50.0/24	.1
99	Management	192.168.99.0/24	.1
999	Native/Blackhole	unused	n/a

Phases and Deliverables

Phase 1: Design and Planning

- Draw the logical and physical topology diagrams
- Build the VLAN/IP/port assignment table
- **Deliverable:** Network design document (diagram + addressing table)

Phase 2: Layer 2 Build

- Create VLANs, assign access ports, configure 802.1Q trunks
- Set the native VLAN to 999, and prune allowed VLANs on trunks
- Configure VTP transparent (or disabled) and verify
- **Deliverable:** Switch configs + `show vlan brief` / `show interfaces trunk` screenshots

Phase 3: Inter-VLAN Routing

- Router-on-a-stick with subinterfaces (`encapsulation dot1Q`)
- Configure DHCP pools per VLAN with excluded addresses and DHCP relay if needed
- Test with ping and traceroute across VLANs
- **Deliverable:** Router config + connectivity test matrix

Phase 4: Upgrade to Layer 3 Switch

- Replace router-on-a-stick with SVIs and `ip routing`
- Compare the two designs (performance, scalability, cost)
- **Deliverable:** Short comparison write-up (half a page)

Phase 5: Routing Extension

- Add a second site or ISP router
- Configure a default route, static routes, then OSPF single-area
- **Deliverable:** Routing tables + `show ip ospf neighbor` evidence

Phase 6: Security and Access Control

- Extended ACLs: Guest can reach the internet but not internal VLANs; HR is isolated from Sales
- Port security, BPDU guard, and shutdown of unused ports
- SSH-only management, disabled Telnet, and encrypted passwords

- **Deliverable:** ACL policy table + before/after test results

Phase 7: Troubleshooting Lab

Inject and fix 5 faults:

1. Wrong access VLAN on a port
2. Trunk native VLAN mismatch
3. VLAN missing from the allowed trunk list
4. Wrong subinterface encapsulation
5. ACL blocking legitimate traffic

Deliverable: Troubleshooting log (symptom, diagnosis commands, root cause, fix)

Phase 8: Documentation and Presentation

- Final report, config backups, and a 5-minute walkthrough demo
- **Deliverable:** PDF report + `.pkt` /GNS3 project file + demo recording

Final Deliverables Checklist

1. Topology diagrams (logical and physical)
2. IP and VLAN addressing plan
3. Full device configurations (text files)
4. Test plan with pass/fail results
5. Security policy and ACL documentation
6. Troubleshooting log
7. Final report (design decisions, lessons learned)
8. Saved simulation file and demo video

Grading Rubric (self-assessment)

Area	Weight
Correct VLAN/trunk setup	20%
Inter-VLAN routing and DHCP working	20%
Routing protocol extension	15%
Security and ACLs	15%

Area	Weight
Troubleshooting skill	15%
Documentation quality	15%

Stretch Goals

- EtherChannel (LACP) between switches
- HSRP for gateway redundancy
- Spanning tree tuning (root bridge, PortFast, RSTP)
- Automate config generation with Python (Netmiko/Jinja2)